MIT 6.1200J
Problem Set 4

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Problem 1: Asymptotic Relationships

(a) Table of asymptotic relationships (no justification required):

$f(n)$	$g(n)$	Relationship
6^n	n^{1200}	$f \in \omega(g)$
$(\log n)^6$	$\sqrt[1200]{n}$	$f \in o(g)$
$\log_6(n^{1200})$	$(\log_{1200} n)^6$	$f \in o(g)$
$\log(n!)$	$n \log n$	$f \sim g$
$2^{\lceil \log_2 n \rceil}$	$3^{\lceil \log_3 n \rceil}$	$f \in \Theta(g)$
$2^{\lceil \log_2 n \rceil} - n + 1$	$3^{\lceil \log_3 n \rceil} - n + 1$	None
n	$n^{\cos \log n}$	$f \in \Omega(g)$

Brief notes on selected rows:

- $\log_6(n^{1200}) = \frac{1200}{\ln 6} \ln n = \Theta(\log n)$, while $(\log_{1200} n)^6 = \Theta((\log n)^6)$. Since $\log n \in o((\log n)^6)$, we get $f \in o(g)$.
- $\log(n!) \sim n \log n$ by Stirling's approximation.
- For $2^{\lceil \log_2 n \rceil}$: since $\log_2 n \leq \lceil \log_2 n \rceil < \log_2 n + 1$, we get $n \leq 2^{\lceil \log_2 n \rceil} < 2n$, so $2^{\lceil \log_2 n \rceil} = \Theta(n)$. Similarly $3^{\lceil \log_3 n \rceil} = \Theta(n)$, so both are $\Theta(n)$ and the ratio is bounded: $f \in \Theta(g)$.
- For the fifth row minus $n + 1$: both $2^{\lceil \log_2 n \rceil} - n + 1$ and $3^{\lceil \log_3 n \rceil} - n + 1$ are $O(n)$ but can oscillate between $\Theta(1)$ and $\Theta(n)$, so no clean relationship holds.

(b) Justified pairs:

Pair 1: $f(n) = 2^{((\log n)^2)}$, $g(n) = 2^{(\lceil \log n \rceil^2)}$
Relationship: $f \in O(g)$

Justification:
By the definition of the ceiling function, we know that for any positive integer $n \geq 1$, $\log n \leq \lceil \log n \rceil$. Assuming the base of the logarithm is 2 (as specified in the problem instructions), $\log n \geq 0$ for $n \geq 1$. This allows us to square both sides while preserving the inequality:

$$(\log n)^2 \leq (\lceil \log n \rceil)^2$$

Because the exponential function 2^x is monotonically increasing, we can exponentiate both sides:

$$2^{(\log n)^2} \leq 2^{(\lceil \log n \rceil)^2}$$

This proves that $f(n) \leq g(n)$ for all $n \geq 1$. Therefore, $f(n)$ is bounded above by $g(n)$, establishing that $f \in O(g)$.

To confirm this is the most specific relationship, we must verify why it is not $o(g)$ or $\Theta(g)$:

- **Why not $o(g)$?** If n is an exact power of 2 (i.e., $n = 2^k$ for some integer k), then $\log_2 n = k$. The ceiling function has no effect, meaning $f(n) = g(n)$. Since $f(n)/g(n) = 1$ for infinitely many values of n, $f(n)$ cannot be strictly asymptotically smaller than $g(n)$.
- **Why not $\Theta(g)$?** If we consider n just below a power of 2, the ceiling function jumps up to the next integer, creating a growing difference between $(\log n)^2$ and $(\lceil \log n \rceil)^2$. Because this gap in the exponent grows unboundedly as n increases, $g(n)$ is not bounded above by $c \cdot f(n)$ for any constant c.

Thus, $O(g)$ is the strongest, most specific relationship.

Pair 2: $f(n) = 4^n$, $g(n) = \frac{(2n)!}{(n!)^2}$ **Relationship:** $f \in \omega(g)$

Justification:
Notice that $g(n)$ is the expanded formula for the central binomial coefficient, $\binom{2n}{n}$. To analyze its asymptotic growth and compare it to $f(n)$, we can apply Stirling's approximation, $n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$:

$$g(n) = \frac{(2n)!}{(n!)^2} \sim \frac{\sqrt{4\pi n}\left(\frac{2n}{e}\right)^{2n}}{\left(\sqrt{2\pi n}\left(\frac{n}{e}\right)^n\right)^2}$$

First, we simplify the denominator:

$$\left(\sqrt{2\pi n}\left(\frac{n}{e}\right)^n\right)^2 = 2\pi n\left(\frac{n}{e}\right)^{2n} = 2\pi n \cdot \frac{n^{2n}}{e^{2n}}$$

Now substituting this back into the approximation for $g(n)$:

$$g(n) \sim \frac{2\sqrt{\pi n} \cdot \frac{2^{2n} n^{2n}}{e^{2n}}}{2\pi n \cdot \frac{n^{2n}}{e^{2n}}}$$

Canceling out the common terms (n^{2n} and e^{2n}) yields:

$$g(n) \sim \frac{2 \cdot 4^n \sqrt{\pi n}}{2\pi n} = \frac{4^n}{\sqrt{\pi n}}$$

Now, to compare $f(n)$ and $g(n)$, we evaluate the limit of their ratio as $n \to \infty$:

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \lim_{n\to\infty} \frac{4^n}{\left(\frac{4^n}{\sqrt{\pi n}}\right)} = \lim_{n\to\infty} \sqrt{\pi n} = \infty$$

Because the limit of the ratio $\frac{f(n)}{g(n)}$ diverges to infinity, it demonstrates that $f(n)$ grows strictly faster asymptotically than $g(n)$. By definition, this establishes the Little-omega relationship: $f \in \omega(g)$.

Problem 2: OEIS

(a) Compute x_3, x_4, x_5, x_6 using $x_n = 5x_{n-2} + 2x_{n-3}$:

$$\begin{aligned} x_3 &= 5x_1 + 2x_0 = 5(1) + 2(0) = 5, \\ x_4 &= 5x_2 + 2x_1 = 5(2) + 2(1) = 12, \\ x_5 &= 5x_3 + 2x_2 = 5(5) + 2(2) = 29, \\ x_6 &= 5x_4 + 2x_3 = 5(12) + 2(5) = 70. \end{aligned}$$

(b) Searching OEIS for $0, 1, 2, 5, 12, 29, 70$ leads to **OEIS A000129**, which matches this sequence.

(c) From the OEIS entry, a closed form is:

$$x_n = \frac{(1+\sqrt{2})^n - (1-\sqrt{2})^n}{2\sqrt{2}}.$$

(d) **Proof by Strong Induction:**

Let $\alpha = 1 + \sqrt{2}$ and $\beta = 1 - \sqrt{2}$. We want to prove that the closed-form formula:

$$x_n = \frac{\alpha^n - \beta^n}{2\sqrt{2}}$$

holds for all integers $n \geq 0$.

Base Cases: Since our recurrence relation $x_n = 5x_{n-2} + 2x_{n-3}$ relies on the three previous terms, we must verify the formula for our first three terms ($n = 0, 1$, and 2):

$$\begin{aligned} n = 0: &\quad \frac{(1+\sqrt{2})^0 - (1-\sqrt{2})^0}{2\sqrt{2}} = \frac{1-1}{2\sqrt{2}} = 0 = x_0. \quad \checkmark \\ n = 1: &\quad \frac{(1+\sqrt{2})^1 - (1-\sqrt{2})^1}{2\sqrt{2}} = \frac{1+\sqrt{2}-1+\sqrt{2}}{2\sqrt{2}} = \frac{2\sqrt{2}}{2\sqrt{2}} = 1 = x_1. \quad \checkmark \\ n = 2: &\quad \frac{(1+\sqrt{2})^2 - (1-\sqrt{2})^2}{2\sqrt{2}} = \frac{(3+2\sqrt{2}) - (3-2\sqrt{2})}{2\sqrt{2}} = \frac{4\sqrt{2}}{2\sqrt{2}} = 2 = x_2. \quad \checkmark \end{aligned}$$

Inductive Hypothesis: Assume that the formula holds for all integers m such that $0 \leq m \leq k$ for some integer $k \geq 2$.

Inductive Step: We want to show that the formula holds for $n = k + 1$. Using the given recurrence relation and our inductive hypothesis:

$$\begin{aligned} x_{k+1} &= 5x_{k-1} + 2x_{k-2} \\ &= 5\left(\frac{\alpha^{k-1} - \beta^{k-1}}{2\sqrt{2}}\right) + 2\left(\frac{\alpha^{k-2} - \beta^{k-2}}{2\sqrt{2}}\right) \\ &= \frac{1}{2\sqrt{2}}\left[\left(5\alpha^{k-1} + 2\alpha^{k-2}\right) - \left(5\beta^{k-1} + 2\beta^{k-2}\right)\right] \\ &= \frac{1}{2\sqrt{2}}\left[\alpha^{k-2}(5\alpha + 2) - \beta^{k-2}(5\beta + 2)\right] \end{aligned}$$

Now, we evaluate $5\alpha + 2$ and compare it to α^3:

$$5\alpha + 2 = 5(1+\sqrt{2}) + 2 = 7 + 5\sqrt{2}$$
$$\alpha^3 = (1+\sqrt{2})^3 = 1^3 + 3(1)^2(\sqrt{2}) + 3(1)(\sqrt{2})^2 + (\sqrt{2})^3 = 1 + 3\sqrt{2} + 6 + 2\sqrt{2} = 7 + 5\sqrt{2}$$

Thus, $5\alpha + 2 = \alpha^3$. By identical algebraic expansion for the conjugate, $5\beta + 2 = \beta^3$.

Substituting these cubic identities back into our equation for x_{k+1}:

$$\begin{aligned} x_{k+1} &= \frac{1}{2\sqrt{2}}\left[\alpha^{k-2}(\alpha^3) - \beta^{k-2}(\beta^3)\right] \\ &= \frac{\alpha^{k+1} - \beta^{k+1}}{2\sqrt{2}} \end{aligned}$$

This matches our target formula for $n = k + 1$. By the principle of strong induction, the formula holds for all integers $n \geq 0$. ■

Problem 3: Bounded Asymptotics

(a) Prove that $f \in O(g)$
Premise: f is bounded from above, and g is bounded away from zero. Both functions map $\mathbb{N} \to \mathbb{R}^+$.

Proof:
Because f is bounded from above, there exists some real number M such that $f(n) \leq M$ for all $n \in \mathbb{N}$. Since the codomain of f is $\mathbb{R}^+$, we know $f(n) > 0$, which means $M > 0$. Because g is bounded away from zero, there exists some real number $B > 0$ such that $g(n) \geq B$ for all $n \in \mathbb{N}$. By the formal definition of Big-O, we need to show that there exist constants $c > 0$ and $n_0 \geq 1$ such that $f(n) \leq c \cdot g(n)$ for all $n \geq n_0$. Let $c = \frac{M}{B}$. Since $M > 0$ and $B > 0$, the constant c is strictly greater than zero. Because $g(n) \geq B$, we can multiply both sides by c to get:

$$c \cdot g(n) \geq c \cdot B = \left(\frac{M}{B}\right) B = M$$

We already established that $f(n) \leq M$. Therefore, by transitivity:

$$f(n) \leq M \leq c \cdot g(n)$$

This inequality holds for all $n \in \mathbb{N}$ (so we can choose $n_0 = 1$). Thus, $f \in O(g)$. ■

(b) Is it necessarily true that $f \in O(g)$?
Answer: No, it is not necessarily true.

Justification:
We can prove this by providing a counterexample. The problem defines "bounded from below" as simply $g(n) \geq B$ for some real number B. It does *not* specify that $B > 0$ (unlike "bounded away from zero").

- Let $f(n) = 1$. This maps to $\mathbb{R}^+$ and is bounded from above (e.g., by $M = 1$).
- Let $g(n) = \frac{1}{n}$. This maps to $\mathbb{R}^+$ and is bounded from below (e.g., by $B = 0$), but it approaches zero as n grows.

If $f \in O(g)$ were true, there would exist constants $c > 0$ and n_0 such that for all $n \geq n_0$:

$$1 \leq c \cdot \left(\frac{1}{n}\right) \implies n \leq c$$

This inequality is impossible to satisfy for all $n \geq n_0$ because c is a fixed constant, but n grows infinitely large. Therefore, $f \notin O(g)$.

(c) Strictly Decreasing and Increasing Functions

Part 1: Is it necessarily true that $f \in O(g)$?
Answer: Yes.

Justification:

- Because f is strictly decreasing and maps $\mathbb{N} \to \mathbb{R}^+$, its largest value is its first term. Thus, $f(n) \leq f(1)$ for all $n \geq 1$. This means f is bounded from above by $M = f(1)$.
- Because g is strictly increasing and maps $\mathbb{N} \to \mathbb{R}^+$, its smallest value is its first term. Thus, $g(n) \geq g(1)$ for all $n \geq 1$. Since g maps to strictly positive real numbers, $g(1) > 0$. Therefore, g is bounded away from zero by $B = g(1)$.
- Because f is bounded from above and g is bounded away from zero, we can apply the exact proof from **Part (a)** to conclude that $f \in O(g)$ must be true.

Part 2: Is it necessarily true that $f \in o(g)$?
Answer: No.

Justification:
For Little-o to hold, the limit $\lim_{n\to\infty} \frac{f(n)}{g(n)}$ must equal 0. We can provide a counterexample where f strictly decreases to a positive constant and g strictly increases to a positive constant:

- Let $f(n) = 1 + \frac{1}{n}$. (Strictly decreasing, maps to $\mathbb{R}^+$).
- Let $g(n) = 2 - \frac{1}{n}$. (Strictly increasing, maps to $\mathbb{R}^+$).

Taking the limit of their ratio as n approaches infinity:

$$\lim_{n\to\infty} \frac{1 + \frac{1}{n}}{2 - \frac{1}{n}} = \frac{1}{2}$$

Because the limit is $\frac{1}{2}$ and not 0, f does not grow strictly slower than g. Therefore, $f \notin o(g)$.

Problem 4: Karatsuba's Algorithm

(a) Compute $T(2)$, $T(3)$, $T(4)$ using $T(n) = 3T(\lceil n/2 \rceil) + 10n$, $T(0) = 0$, $T(1) = 1$:

$$T(2) = 3T(\lceil 2/2 \rceil) + 10(2) = 3T(1) + 20 = 3(1) + 20 = 23,$$
$$T(3) = 3T(\lceil 3/2 \rceil) + 10(3) = 3T(2) + 30 = 3(23) + 30 = 69 + 30 = 99,$$
$$T(4) = 3T(\lceil 4/2 \rceil) + 10(4) = 3T(2) + 40 = 3(23) + 40 = 69 + 40 = 109.$$

(b) Plug and Chug for $n = 2^k$:

Let $n = 2^k$. Then $\lceil n/2 \rceil = 2^{k-1}$, so the recurrence becomes $T(2^k) = 3T(2^{k-1}) + 10 \cdot 2^k$.

Let $S(k) = T(2^k)$. Then $S(k) = 3S(k-1) + 10 \cdot 2^k$ with $S(0) = T(1) = 1$.

Unrolling:

$$\begin{aligned} S(k) &= 3S(k-1) + 10 \cdot 2^k \\ &= 3\big(3S(k-2) + 10 \cdot 2^{k-1}\big) + 10 \cdot 2^k = 3^2 S(k-2) + 3 \cdot 10 \cdot 2^{k-1} + 10 \cdot 2^k \\ &= 3^2 S(k-2) + 10 \cdot 2^k \left(\frac{3}{2} + 1\right) \\ &= 3^j S(k-j) + 10 \cdot 2^k \sum_{i=0}^{j-1} \left(\frac{3}{2}\right)^i. \end{aligned}$$

Setting $j = k$:

$$S(k) = 3^k S(0) + 10 \cdot 2^k \sum_{i=0}^{k-1} \left(\frac{3}{2}\right)^i = 3^k + 10 \cdot 2^k \cdot \frac{(3/2)^k - 1}{3/2 - 1} = 3^k + 10 \cdot 2^k \cdot \frac{(3/2)^k - 1}{1/2}.$$

$$= 3^k + 20 \cdot 2^k \cdot \left(\frac{3^k}{2^k} - 1\right) = 3^k + 20 \cdot 3^k - 20 \cdot 2^k = 21 \cdot 3^k - 20 \cdot 2^k.$$

Since $n = 2^k$, we have $k = \log_2 n$, $3^k = 3^{\log_2 n} = n^{\log_2 3}$, and $2^k = n$. So:

$$\boxed{T(n) = 21\, n^{\log_2 3} - 20n, \quad n \text{ a power of 2.}}$$

Verification:

$$\begin{aligned} T(1): &\quad 21 \cdot 1 - 20 \cdot 1 = 1. \checkmark \\ T(2): &\quad 21 \cdot 3^1 - 20 \cdot 2 = 63 - 40 = 23. \checkmark \\ T(4): &\quad 21 \cdot 3^2 - 20 \cdot 4 = 189 - 80 = 109. \checkmark \end{aligned}$$

(c) Master Theorem application.

The recurrence is $T(n) = 3T(\lceil n/2 \rceil) + 10n$ for $n \geq 2$.

This fits the Master Theorem with $a = 3$, $b = 2$, $f(n) = 10n$.

Compute $n^{\log_b a} = n^{\log_2 3} \approx n^{1.585}$.

Compare $f(n) = 10n$ with $n^{\log_2 3}$: since $\log_2 3 > 1$, we have $n \in o(n^{\log_2 3})$, i.e., $f(n) \in O(n^{\log_2 3 - \epsilon})$ for $\epsilon = \log_2 3 - 1 > 0$.

This satisfies Case 1 of the Master Theorem ($f(n) \in O(n^{\log_b a - \epsilon})$ for some $\epsilon > 0$).

Conclusion:

$$T(n) \in \Theta(n^{\log_2 3}).$$

Proof using Master Theorem: We have $a = 3 \geq 1$, $b = 2 > 1$, $f(n) = 10n = \Theta(n^1)$. Since $\log_2 3 \approx 1.585 > 1$, we have $f(n) = O(n^{\log_2 3 - \epsilon})$ for $\epsilon = \log_2 3 - 1 > 0$. By Case 1 of the Master Theorem, $T(n) \in \Theta(n^{\log_2 3})$. ■

(d) Is Karatsuba an asymptotic improvement?

Yes. The straightforward algorithm runs in $\Theta(n^2)$, while Karatsuba runs in $\Theta(n^{\log_2 3})$. Since $\log_2 3 \approx 1.585 < 2$, we have $n^{\log_2 3} \in o(n^2)$, so Karatsuba is **asymptotically faster** than the straightforward algorithm.